7.6 BERNOULLI'S THEOREM

7.6.1 Equation of continuity

In order to discuss the mass flow rate through a pipe, it is necessary to assume that the flow of fluid is steady, the flow of the fluid is said to be steady if at any given point, the velocity of each passing fluid particle remains constant with respect to time. Under this condition, the path taken by the fluid particle is a streamline.

Consider a pipe AB of varying cross sectional area a_1 and a_2 such that $a_1 > a_2$. A non-viscous and incompressible liquid flows steadily through the pipe, with velocities v_1 and v_2 in area a_1 and a_2, respectively as shown in Figure 7.32.

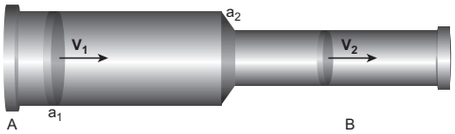


Fig 7.32 A streamlined flow of fluid through a pipe of varying cross sectional area

Let m_1 be the mass of fluid flowing through section A in time Δt, $m_1 = (a_1 v_1 \Delta t)\ \rho$

Let m_2 be the mass of fluid flowing through section B in time Δt, $m_2 = (a_2 v_2 \Delta t)\ \rho$

For an incompressible liquid, mass is conserved $m_1 = m_2$

$$a_1 v_1\ \Delta t\ \rho = a_2 v_2 \Delta t\ \rho$$

$$a_1 v_1 = a_2 v_2 \Rightarrow a\ v = \text{constant} \qquad (7.35)$$

which is called the equation of continuity and it is a statement of conservation of mass in the flow of fluids.

In general, $a\ v$ = constant, which means that the volume flux or flow rate remains constant throughout the pipe. In other words, the smaller the cross section, greater will be the velocity of the fluid.

EXAMPLE 7.14

In a normal adult, the average speed of the blood through the aorta (radius $r = 0.8$ cm) is 0.33 ms^{-1}. From the aorta, the blood goes into major arteries, which are 30 in number, each of radius 0.4 cm. Calculate the speed of the blood through the arteries.

Solution:

$a_1 v_1 = 30\ a_2\ v_2 \Rightarrow \pi r_1^2 v_1 = 30\ \pi r_2^2 v_2$

$$v_2 = \frac{1}{30}\left(\frac{r_1}{r_2}\right)^2 v_1$$

$$\Rightarrow v_2 = \frac{1}{30} \times \left(\frac{0.8\times10^{-2}\,\text{m}}{0.4\times10^{-2}\,\text{m}}\right)^2 \times (0.33\,\text{ms}^{-1})$$

$v_2 = 0.044\ \text{m s}^{-1}$

7.6.2 Pressure, kinetic and potential energy of liquids

A liquid in a steady flow can possess three kinds of energy. They are (1) Kinetic energy, (2) Potential energy, and (3) Pressure energy, respectively.

i) **Kinetic energy:** The kinetic energy of a liquid of mass m moving with a velocity v is given by

$$KE = \frac{1}{2} mv^2$$

The kinetic energy per unit mass =

$$\frac{KE}{m} = \frac{\frac{1}{2}mv^2}{m} = \frac{1}{2} v^2$$

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